MASTER FUNCTIONAL SKILLS

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NCFE · LEVEL 2 · MATHEMATICS

NCFE Functional Skills Mathematics Level 2

September 2020 Section B · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: Past-Paper-L2-Functional-Skills-Maths_P001372_QP-B

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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Activity 2: Police workshop

2 (a) Alice lives in Oldcastle.

She is on a Community Policing course.

The ratio of solved to unsolved crimes in Oldcastle is 2 : 9

$\frac{1}{8}$ of crimes are solved in the whole of the UK.

Alice thinks that a greater proportion of crimes are solved in Oldcastle than in the rest of the UK.

Is she correct?

Show how you decide.

[3 marks]

Q2(a) Worked Solution:

See below

Alternative method 1 (fractions)

1

2

11

1

16

88 AND 11

88

OR 0.18(..) AND 0.125

[3 marks]

Your answer:

PAST PAPER

Please turn over

Examiner use only

2 (b) The value of items stolen in Oldcastle has increased by 8% each year.

The value of stolen items in 2017 was £1.7 million.

What was the value of items stolen in 2019?

[2 marks]

Q2(b) ~~Worked Solution~~:
(£)1.98(288) or 1982880
Award 2 marks if correct answer given
1
1.7 × (1.08)2
or 1 700 000 × (1.08)2
OE any full correct method
1
(£)1.98(288) or 1982880
[2 marks]

Your answer: £

PAST PAPER

Examiner use only

2 (c) Alice attends a workshop about home security.

The tutor tells the group about a house burglary.

The police found a footprint in the garden.

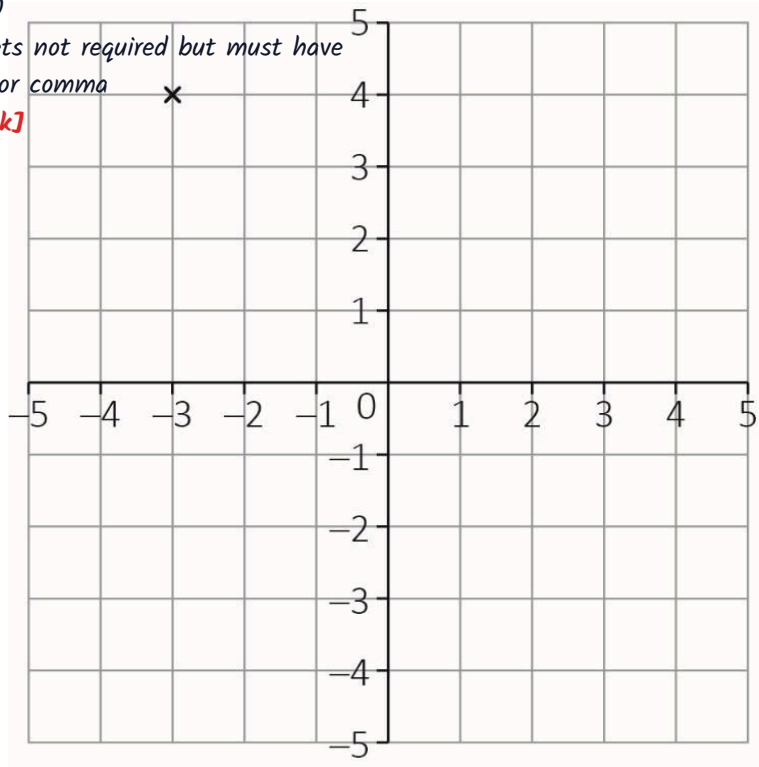
They used a coordinate grid to record where it was:

Q2(c) Worked Solution:

(−3,4)

Brackets not required but must have space or comma

[1 mark]



What are the coordinates of the point marked by ×?

[1 mark]

Your answer:

Please turn over

Examiner use only

2 (d) A witness said the suspect is under 5 feet 10 inches tall, and has blonde hair.

Alice is shown 15 photographs:

		Hair colour	
		Blonde	**Not blonde**
Height	**Under 5 feet 10 inches**	3	6
	Over 5 feet 10 inches	4	2

Q2(d) Worked Solution:

Alice picks one of the photographs at random.

What is the probability that it is of a person who has blonde hair and is under 5 feet 10 inches tall?

See below
1
3
15 or 1
5
H26
1
0.2
FT their fraction
Accept 0.3` only if indicated as a
[2 marks]

Give your answer as:

- a fraction
- a decimal.

[2 marks]

Your answer:

Fraction:
Decimal:

PAST PAPER

Examiner use only

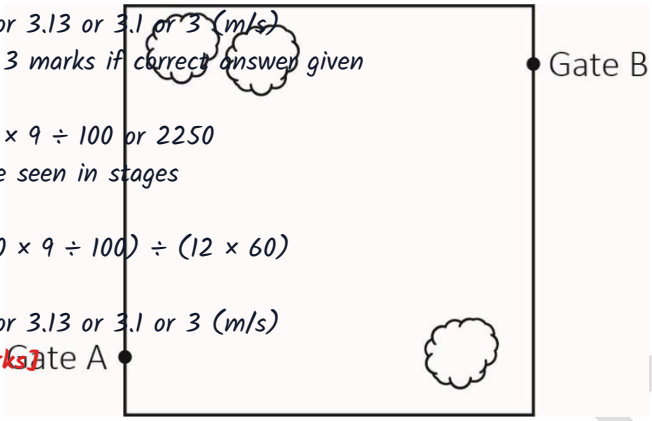
2 (e) At 20:40 a witness saw the suspect entering the park at Gate A.

Another witness saw the same person leaving the park through Gate B exactly 12 minutes later.

The group are given this map of the park:

Q2(e) Worked Solution:

3.125 or 3.13 or 3.1 or 3 (m/s)
Award 3 marks if correct answer given
1
25000 × 9 ÷ 100 or 2250
May be seen in stages
1
(25000 × 9 ÷ 100) ÷ (12 × 60)
1
3.125 or 3.13 or 3.1 or 3 (m/s)
[3 marks]



Not drawn accurately

The map has a scale of 1 : 25 000

The distance between Gates A and B on the map is 9 cm

What was the average speed of the person seen entering and leaving the park?

Give your answer in metres per second (m/s).

[3 marks]

Your answer: **m/s**

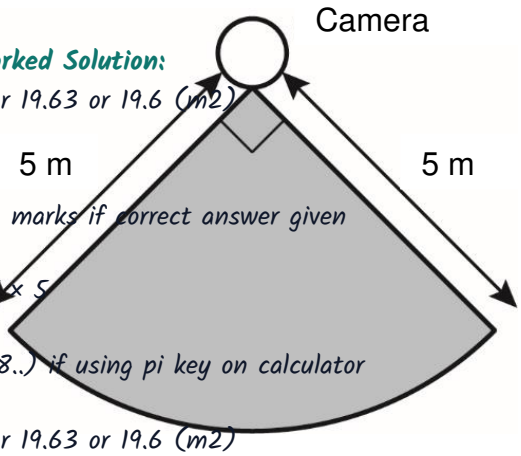
Please turn over

PAST PAPER

Examiner use only

2 (f) The house had a CCTV camera.

The shaded section of the diagram shows the area covered by the camera:



Not drawn accurately

Q2(f) Worked Solution:
19.625 or 19.63 or 19.6 (m2)

Award 2 marks if correct answer given
1
3.14 × 5 × 5
or 78.5
78.53(98..) if using pi key on calculator
1
19.625 or 19.63 or 19.6 (m2)
Accept 19.6(3495408) if using pi key

Calculate the area covered by the camera.

[2 marks]

Use $\pi = 3.14$

[2 marks]

Your answer: **m²**

PAST PAPER

Examiner use only

2 (g) Alice has a digital picture of the burglar from the CCTV camera.

The size of the print is $5\frac{1}{2}$ inches by $3\frac{3}{4}$ inches.

Q2(g) Worked Solution:

See below

Alternative method 1 (cm)

1

5.5 × 2.54 or 13.97

OR 3.75 × 2.54 or 9.525

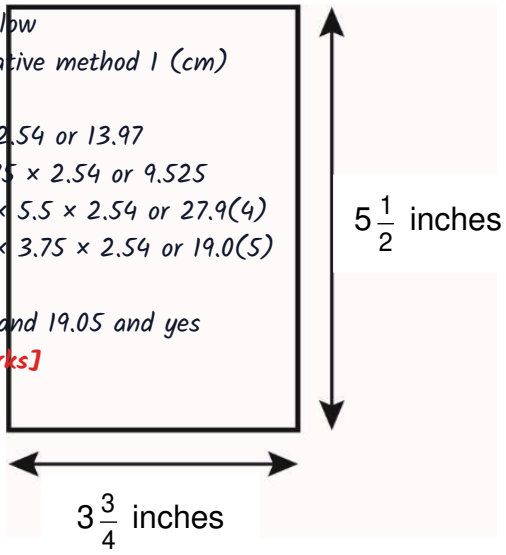
OR 2 × 5.5 × 2.54 or 27.9(4)

OR 2 × 3.75 × 2.54 or 19.0(5)

1

27.94 and 19.05 and yes

[2 marks]



Not drawn accurately

Alice enlarges the picture by doubling both the width and the length.

She wants to print the enlarged picture onto one side of A4 paper.

A4 paper is 21 cm wide and 29.7 cm long.

Will the enlarged picture fit onto one side of A4 paper?

Show how you decide.

Use the conversion: 1 inch = 2.54 cm

[2 marks]

Your answer:

[Total marks: 15]

PAST PAPER

Examiner use only

Activity 3: Volunteering

3 (a) Sam is a volunteer for CHAD, a charity that supports children.

Charities are put into categories according to the amount of money they raise each year.

Last year, CHAD raised five hundred and sixty-nine thousand, eight hundred and one pounds.

Q3(a) Worked Solution:

Tick the box next to the category that CHAD is in.

[1 mark]

Medium
Indicated in any way
[1 mark]

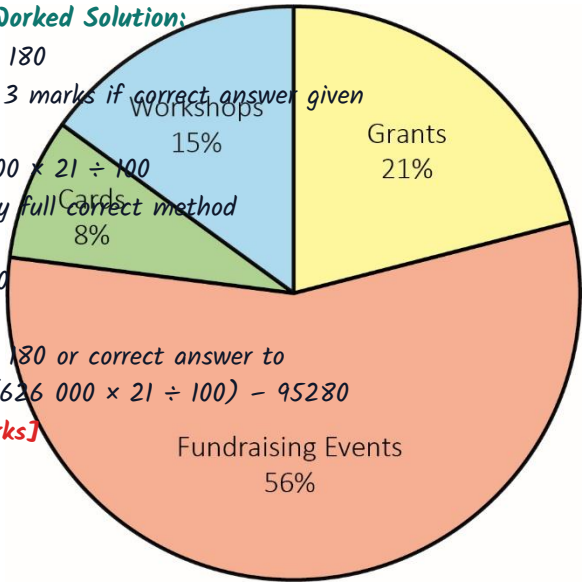
	Category	Money raised each year (£)	
		Minimum	**Maximum**
☐	Micro	0	9 999
☐	Small	10 000	99 999
☐	Medium	100 000	999 999
☐	Large	1 000 000	9 999 999
☐	Major	10 000 000	99 999 999
☐	Super-major	100 000 000	no maximum

PAST PAPER

Examiner use only

3 (b) This year, CHAD aim to raise £626 000

The pie chart shows where the funds will come from:



Q3(b) Worked Solution:
(£)36 180
Award 3 marks if correct answer given
1
626 000 × 21 ÷ 100
OE Any full correct method
1
131 460
1
(£)36 180 or correct answer to their (626 000 × 21 ÷ 100) − 95280
[3 marks]

So far, CHAD have raised £95 280 from grants.

How much more money do they aim to raise from grants this year?

[3 marks]

Your answer: £

PAST PAPER

Please turn over

Examiner use only

3 (c) Sam asks friends and family to sponsor him for a 10 km run.

A local business offers to donate 25p for every £10 that Sam raises.

In total, Sam raised £2634.25

This included the money that the business donated.

Q3(c) Worked Solution:

How much did the business donate to Sam?

[3 marks]

(£)64.25
Award 3 marks if correct answer given
1
2634.25 ÷ 10.25 or 257
1
(2634.25 ÷ 10.25) × 0.25
or 2634.25 – (2634.25 ÷ 10.25 × 10)
OE any full correct method
1
(£)64.25
[3 marks]

Your answer: **£**

PAST PAPER

Examiner use only

3 (d) CHAD are sending a letter to supporters.

They have asked for volunteers to put the letters into envelopes.

Eight people volunteer to help.

Each person works at the same rate.

Q3(d) Worked Solution:

It takes them 480 minutes to complete the task.

How many minutes would the task have taken 12 people, if each person worked at the same rate?

[2 marks]

320 (minutes)
Award ~~2 marks if correct answer given~~
1
8 × 480 ÷ 12
1
320 (minutes)
[2 marks]

Your answer: **minutes**

PAST PAPER

Please turn over

Examiner use only

3 (e) Some people donate to CHAD by giving a one-off donation.

The table below shows the one-off donations made last year:

Size of one-off donation (£d)	Number of donors
0 < d ≤ 20	48
20 < d ≤ 40	146
40 < d ≤ 60	74
60 < d ≤ 80	32
Total:	**300**

Q3(e) Worked Solution:
36 and 50 AND no
Award 4 marks if correct answer given
1
(10 × 48) + (30 × 146) + (50 × 74) +

Other people donate by paying regular amounts.

Last year, CHAD had 150 of these regular donors who donated £7500 in total.

Sam works out an estimate for the mean of the one-off donations last year.

He compares this with the mean for regular donors last year.

Sam says,

"The mean amount donated last year was higher for the one-off donors than for the regular donors".

PAST PAPER

Question 3(e) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

(70 × 32)

or 480 + 4380 + 3700 + 2240

or 10800

H24

1

Their ((10 × 48) + (30 × 146) + (50 × 74)

[4 marks]

Examiner use only

Is Sam correct?

Show how you decide.

[4 marks]

Your answer:

PAST PAPER

Please turn over

Examiner use only

3 (f) Kay and Mo are employed by CHAD as fundraisers.

They are given targets:

- Kay must raise funds of at least $3\frac{2}{5}$ times her salary.
- Mo must raise funds of at least $\frac{10}{3}$ times his salary.

Which is larger, $3\frac{2}{5}$ or $\frac{10}{3}$?

Show how you decide.

[2 marks]

Your answer:

[Total marks: 15]

Q3(f) Worked Solution:

3 2

5 with working that allows direct comparison

Award 2 marks if correct answer given

1

S1

15 and 50

15

OR 3 6

15 and 3 5

[2 marks]

PAST PAPER

Examiner use only

Activity 4: Selling cars

4 (a) Emma sells cars.

The original price of a car was £15 400

Its sale price is £13 500

Calculate the percentage discount.

Give your answer to one decimal place.

[2 marks]

Q4(a) Worked Solution:

12.3(%)

~~Award 2 marks if correct answer given~~

1

(15 400 – 13 500) × 100 ÷ 15 400

or 1900 × 100 ÷ 15 400

or 12.33(...)

OE full correct method

1

12.3(%)

[2 marks]

Your answer: %

PAST PAPER

Please turn over

Examiner use only

4 (b) Emma also buys cars.

Bill wants to sell his old car.

The car is 3 years old and in good condition.

Emma uses a formula to work out the price that she will pay for Bill's car:

Q4(b) Worked Solution:
See below
1

$$P = 12\,000 \times C \times (0.75)^T$$

Where:

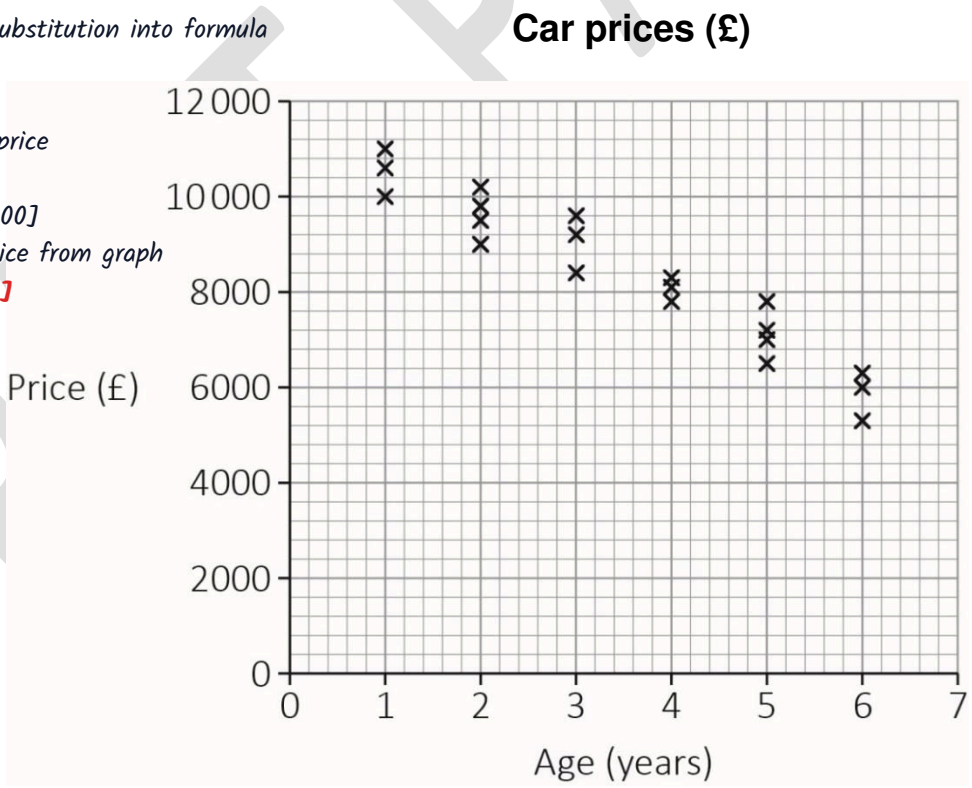
P is the price the she will pay (in £)
T is the age of the car (in years)
C is the condition of the used car, according to this table:

12 000 × 0.9 × (0.75)3

Condition	C
Good	0.9
Average	0.7
Poor	0.5

The scatter diagram below shows the prices paid online for the same model of car in good condition:

Correct substitution into formula
1
4556.25
Emma's price
1
[8400,9600]
Online price from graph
[4 marks]



PRACTICE PAPER

Examiner use only

How much more would Bill expect to get if he sells his car online instead of selling to Emma?

[4 marks]

Your answer: **£**

PAST PAPER

Please turn over

Examiner use only

4 (c) Bill sees a new car he likes.

The car's fuel consumption is 16 **km** per litre.

Bill drives a total of 45 **miles** each day.

Fuel costs 136.9 pence per litre.

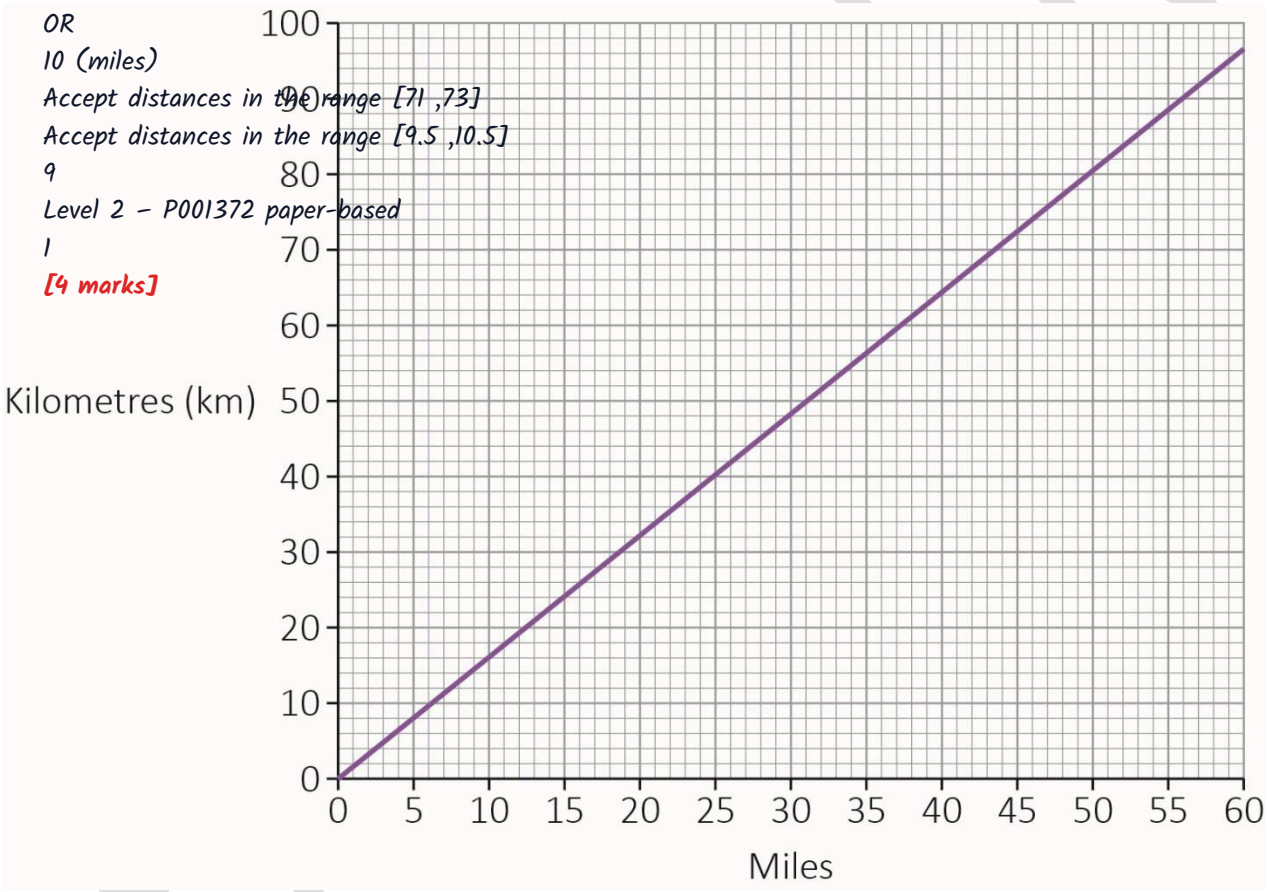
Q4(c) Worked Solution:

Work out the total cost of fuel Bill would use each day in this car.

Give your answer in pounds and to the nearest penny.

[4 marks]

Use this conversion graph to help you:



See below
1
72 (km)
OR
10 (miles)
Accept distances in the range [71 ,73]
Accept distances in the range [9.5 ,10.5]
9
Level 2 – P001372 paper-based
1
[4 marks]

Examiner use only

Your answer: £

PAST PAPER

Please turn over

Examiner use only

4 (d) Emma employs Hans.

She pays Hans a bonus when he sells a car.

The bonus is 1.5% of the price of the car.

The available makes of cars are:

Q4(d) Worked Solution:

32 (cars)

Make of cars	Price
A	£14 750
B	£15 800
C	£16 000
D	£15 800
E	£19 800
F	£15 800
G	£30 500
H	£23 650

Must be supported by calculations
1
15 800
H23b
1
Their 15800 × 1.5 ÷ 100
or
7500 ÷ (1.5 ÷ 100)
Award for any full correct method.
[5 marks]

PAST PAPER

Examiner use only

How many cars would Hans have to sell at the modal price to get a bonus of at least £7500?

You must show your working.

[5 marks]

Your answer: **cars**

PAST PAPER

[Total marks: 15]

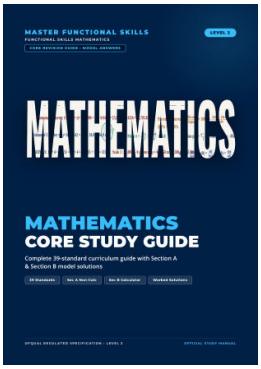
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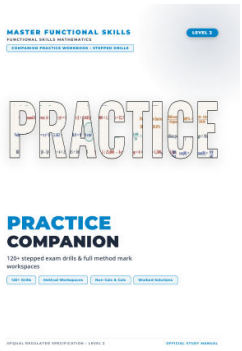


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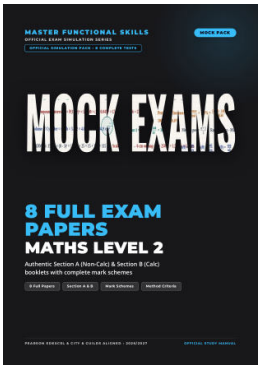


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